

Daniel Morales, Ph.D.

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Languages: Spanish (Native) / English (Certified B2 Cambridge)

SUMMARY

Ph.D. in Computer Science with interest in applied research mainly focused on cybersecurity and applied cryptography (PETs and decentralized environments). Also, solid background in communication networks.

- Background in engineering, focused on Internet and Mobile communication networks and protocols.
- Ph.D. focused on applied cryptography (mainly PETs, with focus on the intersection between MPC and blockchain technology). Understanding of post-quantum cryptography algorithms.
- Experience in working with zk-SNARK/zkVMs designs and libraries (Arkworks, Halo2/Midnight-ZK, Risc Zero, Folding, PQ-secure SNARKs...)
- Experience in teaching aspects, mainly related to cryptography, cybersecurity and blockchain.

EDUCATION

Ph.D., Computer Science (Applied Cryptography), University of Málaga, 2025 (with honors).

Thesis title: *Secure Multi-Party Computation: Constructions and Applications in Blockchain.*

M.Sc., Communication Networks (Mobile networks), University of Málaga, 2020.

B.Sc., Communication Networks (Internet protocols and applications), University of Málaga, 2019.

TECHNICAL EXPERTISE

- Design, configuration and deployment of blockchain infrastructures and applications (Ethereum and Cardano ecosystems).
- Using frameworks and tools to deploy MPC/FHE/ZKP-based protocols and programming of cryptographic libraries.
- Programming experience: C/C++, Rust, Java, NodeJs, Python.

Input Output Global (November 2025 - February 2026)

Applied Cryptographer

Consulting work in applied cryptography with IOG's Applied Research & Creative Engineering team. Support in [Cavefish](#) light client stream: focus on blind signatures and zk-SNARKs (security analysis of cryptographic blocks, benchmarking, prototype support, optimizations design).

Input Output Global (July - October 2025)

Applied Cryptography Internship

Applied research with IOG's innovation team. Focus on SNARK libraries (benchmarks of different gadgets, arithmetization techniques and proof systems). Mainly: Arkworks, Halo 2 and Risc Zero.

RESEARCH EXPERIENCE

Cryptography and Information: Foundational Research and Applications (CIFRA) Lab, Bocconi University, Italy. Supervisor: Emmanuela Orsini (July - September 2024)

Research Internship

Research focused on advanced cryptographic protocols for Secure Multi-Party Computation.

Network, Information and CyberSecurity (NICS) Lab, University of Málaga, Spain (2018-2025)

Predoctoral Researcher

Focused on new constructions and applications to Secure Multi-Party Computation and Blockchain.

SELECTED AWARDS AND GRANTS

- Spanish national competitive grant for Ph.D. students, Ministry of Education, 2019, duration: 4 years
- Best Student Paper Award at 19th International Conference on Security and Cryptography (SECRYPT 2022)
- Best B.Sc. Thesis Award: “MPC-based PKI prototype”, course 2018/2019, University of Málaga

TEACHING EXPERIENCE

- **B.Sc. and M.Sc. level:** e-commerce, cybersecurity, forensics, mobile security... (lectures and lab)
- **Expert course in blockchain technology:** private blockchain and layer 2 solutions (lectures and lab)

SELECTED INTERNATIONAL PUBLICATIONS

- D. Morales, I. Agudo and J. Lopez, “Zero-Knowledge Bitcoin Mixer with Reversible Unlinkability”, in Blockchain: Research and Applications Journal, 2025, DOI: 10.1016/j.bcr.2025.100323
- D. Morales, I. Agudo and J. Lopez, “A Lightweight Mechanism for Dynamic Secret Sharing of Private Data by Constrained Devices”, in IEEE Internet of Things Journal, 2025, DOI: 10.1109/IJOT.2025.3555026
- D. Morales, I. Agudo, J. Lopez, “Toward a Framework for Cost-Effective and Publicly Verifiable Confidential Computations in Blockchain”, in IEEE Communications Magazine, vol. 63, pp. 96-102, 2024, DOI: 10.1109/MCOM.001.2300839
- D. Morales, I. Agudo, J. Lopez, “Integration of MPC into Besu through and extended private transaction model”, in IEEE International Conference on Metaverse Computing, Networking and Applications (METACOM), pp. 266-273, 2023, DOI: 10.1109/MetaCom57706.2023.00056
- D. Morales, I. Agudo, J. Lopez, “Private set intersection: A systematic literature review”, in Computer Science Review, vol. 49, no. 100567, 2023, DOI: 10.1016/j.cosrev.2023.100567
- D. Morales, I. Agudo, J. Lopez, “Real-time Crowd Counting based on Wearable Ephemeral IDs”, in 19th International Conference on Security and Cryptography (SECRYPT 2022), pp. 249-260, 2022, DOI: 10.5220/0011327200003283

ORAL PRESENTATIONS

- D. Morales, I. Agudo, J. Lopez, “Secret Sharing Dealing from Encrypted Data”, in Workshop on secret sharing schemes, 12-14 May 2025, Tarragona, Spain
- D. Morales, I. Agudo, J. Lopez, “Cryptographic mechanisms for Anonymous payments in blockchain”, in Crypto Seminars 2024, Universitat Politècnica de Catalunya, Spain

OTHERS

- Program Committee at the 2nd Workshop on Cryptographic Tools for Blockchains (affiliated with Eurocrypt 2025)
- Organizer at the 3rd Workshop on Cryptographic Tools for Blockchains (affiliated with Eurocrypt 2026)
- Program Committee at the 23rd International Conference on Security and Cryptography (SECRYPT 2026)